Safety Immersion and Gamified Hazard Training (SIGHT)

Participant Forms

VR session

Participant ID

P _____

Thank you for taking part in this study. The research assistant will give you each form when it is time to fill it in. There are no right or wrong answers, and you may leave blank any question you do not want to answer.

Forms in This Packet

Start of Session

#1 Consent Form (about 5 min) . . . 2

#1 Consent Form: your copy to keep . . . 4

#2 Demographic Questionnaire (about 5 min) . . . 6

#3 Job Hazard Analysis Quiz (about 7 min) . . . 7

#4 GATT Questionnaire (about 8 min) . . . 9

End of Session

#2 TAM Questionnaire (about 5 min) . . . 10

DRAFT revision, pending IRB approval

Research Consent Form

MIAMI UNIVERSITY
OXFORD, OH • EST. 1809

Dear participant:

Thank you for considering participation in this study. I am Dr. Jay Shan, an Associate Professor in the Department of Information Systems & Analytics at Miami University. My research team and I are conducting a research project to develop an ecosystem that integrates web-based virtual reality (VR) and augmented reality (AR) for personalized safety training and the goal of this study is to evaluate user feedback of this system.

If you agree to participate, you will be asked to complete 2 sessions, separated by approximately 1 week. Each session will last approximately 2 hours. You will receive a $75 Visa gift card at the end of each session you attend. At the start of the first session, we will ask you to complete a demographic questionnaire and two other questionnaires on hazard evaluation and technology. This should take approximately 20 minutes. We will then ask you to complete one of two training conditions depending on the session. The two conditions are training either with a web-based VR system or with an AR headset system. The VR session will involve a web-based VR scenario presented on a standard 27" monitor where you will be seated and interact with a standard mouse and keyboard to train on a lathe, milling machine, and collaborative robot. For the AR session, you will wear a head-mounted display to complete first person perspective training while standing at a non-operational milling machine, a lathe that runs behind its closed door, and a collaborative robot that moves slowly and stops if it touches anything. The headset shows a zone near the robot that you must not enter.

At the start of the training portion of the session, you will receive a system orientation, where you will learn about using the system. Then, we will ask you to complete three training scenarios. Each of the scenarios will be focused on recognizing and mitigating manufacturing workplace hazards and machine operation. Between scenarios, you will be able to take a break. Following completion of the three scenarios, we will ask you to complete one assessment module per machine (VR only), a post-session feedback questionnaire, and an interview. Your performance on the training scenarios will be recorded.

You will not be asked to include your name on any of the questionnaires, thus your answers cannot be associated with you. The questionnaire responses will be treated as confidential information, stored in a secure location for at least three years after the study, and accessed only by the research team members, under the supervision of Dr. Fadel M. Megahed.

Your questionnaires will be scanned into a password-protected study website. OpenAI, an artificial intelligence service, will help read the handwriting and find themes in the interview answers after names are removed, and a research team member will check every response. With your permission, the interview (approximately 5–10 minutes, 4 questions) will also be recorded and transcribed offline on the SIGHT workstation; otherwise, the researcher will take notes. Recordings will be labeled with your study ID only and deleted once the transcript is checked. According to OpenAI's documentation (https://developers.openai.com/api/docs/guides/your-data), information sent through its API is not used to train its models.

Your participation is voluntary and you may withdraw from the study at any time or decline to answer any questions that make you uncomfortable. If you withdraw, your responses will be retained unless

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SIGN AND RETURN TO THE RESEARCH ASSISTANT

you ask that they be deleted. You will not be asked to do anything that exposes you to risks beyond those of everyday life. During the study, if you feel any discomfort with the use of the VR or AR systems, you may ask to stop the session. The benefit of the study, scientifically, is to help develop a proof-of-concept training system that integrates AI-driven safety insights, gamified web-based VR, and AI-powered AR for real-time safety and productivity guidance. This project aims to provide an immersive, adaptable, and scalable hazard training tool for Ohio's manufacturing workforce. At the end of the study, the researcher will make the general results available to you. The generalized results may be presented at professional conferences or published in articles describing the results of the research. Only aggregate results and unidentified interview quotes will be reported.

When using virtual reality systems, you could experience one or more of the following: disorientation, dizziness, eye strain, headache, lightheadedness, nausea, or sweating. It is possible that these could linger after participation, or that there may be unknown aftereffects not identified by current research. If there is any change in the way you are feeling, please let the researchers know immediately. If you feel any effects, you will be asked to remain at the lab until any side effects have subsided.

If you choose to participate in the study, you may discontinue participation at any time without penalty or loss of benefits. By signing the form, you do not give up any personal legal rights you may have as a participant in this study. You may refuse to participate in this study without penalty of loss of benefits to which you are otherwise entitled.

If you suffer an injury from participation in the study, you should notify the researcher immediately. By signing this form you acknowledge and agree that (1) the cost for medical treatment will be billed to you or your insurance, and (2) Miami University will not be responsible for any medical bills incurred while participating in this study. Miami University has no funds set aside for the payment of health care expenses for this study.

If you have any questions about the research now or in the future you may contact me at jayshan@miamioh.edu or (513) 529-2135. If you have any questions or concerns about the rights of research subjects or the voluntariness of participation, you may contact the Research Ethics and Integrity Program at Miami University at either (513) 529-3600 or humansubjects@miamioh.edu.

Thank you for your participation. We are very grateful for your help and hope that this will be an interesting session for you. You may keep this portion of the page. This project has been reviewed and approved by Miami Institutional Review Board (Level 2): Protocol Number 02415r.

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I agree to participate in this study about VR and AR training for workplace safety. I understand my participation is voluntary and that my name will not be associated with my responses. By signing below, I acknowledge that I am 18 years or older.

Audio recording of my interview (about 5–10 minutes, 4 questions): ☐ I agree ☐ I do not agree

Participant Signature _____ Date: _____

Researcher Signature _____ Date: _____

SIGN AND RETURN TO THE RESEARCH ASSISTANT

DRAFT

DRAFT revision, pending IRB approval

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Audio recording of my interview (about 5–10 minutes, 4 questions): ☐ I agree ☐ I do not agree

Participant Signature _____ Date: _____

Researcher Signature _____ Date: _____

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Participant ID: P _____ This form takes about 5 minutes.

Demographic Information

Please answer each question. You may leave blank any question you do not want to answer.

Age	_____
Gender	☐ Male ☐ Female
Hand dominance	☐ Left ☐ Right ☐ Ambidextrous
Do you wear corrective lenses (glasses or contact lenses)?	☐ Yes ☐ No
If yes, are you wearing them now?	☐ Yes ☐ No
Current job role (e.g., student, machinist, etc.)	_____
Years of experience with machine shop equipment	_____

Prior relevant training for machine safety (choose all that apply)

☐ a) Classroom training (e.g., lectures)

☐ b) Online training (e.g., video-based short course)

☐ c) Hands-on laboratory safety training

☐ d) Other: _____

When did you last complete a machine safety training? (choose one)

☐ a) Never

☐ b) Within the last 3 months

☐ c) Within the last year

☐ d) More than 1 year ago

When did you last use either a manual mill, lathe, or collaborative robot? (choose one)

☐ a) Never

☐ b) Within the last 2 weeks

☐ c) Within the last 3 months

☐ d) Within the last 6 months

☐ e) More than 6 months ago

Do you have any condition, such as vision abnormalities, pregnancy, a heart condition, or illness that includes headache, nausea, or dizziness? ☐ YES ☐ NO

Participant ID: P _____ This quiz takes about 7 minutes.

Job Hazard Analysis Quiz

Write the letter of your answer (a, b, c, or d), or **T** for true or **F** for false, in the box to the left of each question. Choose one answer per question.

☐ **1.** A job hazard analysis is most effective when carried out by:

a) The engineer who designed the equipment or process
b) The manager who oversees the area
c) A team of workers, supervisors, and subject matter experts who all understand the operation
d) A person with specialized JHA training, such as the safety manager

☐ **2.** When you cannot eliminate a hazard, what type of control measure should you use **first** to protect workers from it?

a) Engineering controls
b) Administrative controls
c) Personal protective equipment
d) Safety training

☐ **3.** True or false: The best way to control any potential hazard is to eliminate it entirely. (T or F)

☐ **4.** Which of these could be a source of hazards?

a) Rotating parts on a machine
b) Repetitive lifting of objects
c) Walking on a wet floor
d) All of the above

☐ **5.** Which of these should you discuss when preparing a JHA?

a) Step or task, risk assessment to determine the worst-case injury, person responsible for implementing corrective action
b) Step or task, potential hazards, needed control measures
c) Potential hazards, risk assessment to determine the most likely injury, PPE needed
d) Potential hazards, PPE needed, person responsible for corrective action

☐ **6.** Which of these is **not** an engineering control?

a) A device that blocks the hazard at the source before it comes into contact with the worker
b) A ventilation system to reduce exposure to an air contaminant
c) Gloves
d) Guardrails on a scaffold

Participant ID: P _____ Job Hazard Analysis Quiz, continued

☐ **7.** Which of these is an administrative control?

a) Personal protective equipment
b) A changed procedure
c) An enclosure around a machine to reduce the noise it produces
d) Substituting a chemical in a process with a less hazardous one

☐ **8.** When choosing jobs to conduct JHAs for, you should prioritize:

a) Jobs that might cause severe or disabling injuries or illness, including those with no history of previous incidents
b) Jobs that are new or have undergone changes
c) Jobs that have detailed written instructions because they are complex
d) All of the above

☐ **9.** When should you review an existing JHA?

a) When there is a change in the job or if an injury or illness occurs on a specific job
b) Periodically (to ensure it is still accurate and to find hazards that the initial analysis did not identify)
c) Never
d) Both answers (a) and (b)

☐ **10.** True or false: A JHA is a method for identifying hazards; it examines the steps in the job, then identifies ways to control the hazards found. (T or F)

☐ **11.** Which of these is **not** a benefit of JHAs?

a) They prevent workplace injuries and illnesses
b) They raise workplace productivity
c) They qualify the workplace for an OSHA recognition program
d) They help a worksite comply with OSHA regulations

☐ **12.** Using PPE as a control strategy is acceptable:

a) While engineering controls are being developed and installed
b) When engineering controls are determined to be infeasible or impracticable
c) When all other controls are not enough to reduce the exposure to the hazard
d) All of the above

Source: OSHA Safe + Sound Week 2024, "The Hunt for Hazards" Job Hazard Analysis Quiz.

Participant ID: P _____ This questionnaire takes about 8 minutes.

Attitudes Toward Technology

Please read each statement and mark **one circle** to show how much you agree or disagree with it. There are no right or wrong answers. You may leave blank any statement you do not want to answer.

		1 Strongly disagree	2 Disagree	3 Slightly disagree	4 Slightly agree	5 Agree	6 Strongly agree
1.	Technology makes our lives easier and more comfortable.	○	○	○	○	○	○
2.	Technology can sort out any problem.	○	○	○	○	○	○
3.	Continued technological innovation is essential for improving people's life satisfaction.	○	○	○	○	○	○
4.	Technology makes our way of life change too fast.	○	○	○	○	○	○
5.	Technology expands our limits.	○	○	○	○	○	○
6.	Technology creates novel problems.	○	○	○	○	○	○
7.	Technology can sometimes damage people's moral sense.	○	○	○	○	○	○
8.	The applications of technology can threaten human rights.	○	○	○	○	○	○
9.	New inventions will always be found that counteract any harmful effect of technological developments.	○	○	○	○	○	○
10.	The benefits of technology are greater than any harmful effects it may have.	○	○	○	○	○	○
11.	If a new technology poses risks that are uncertain and not yet fully understood, the development of this technology should be stopped even if benefits are expected.	○	○	○	○	○	○
12.	If we attach too many risks that are not yet fully understood, we could miss out on technological progress.	○	○	○	○	○	○
13.	Limits need to be set for technological progress.	○	○	○	○	○	○
14.	I am worried about new technologies fraught with uncertainties.	○	○	○	○	○	○
15.	Without technological innovation the economy would be less stable.	○	○	○	○	○	○
16.	Technology means progress.	○	○	○	○	○	○
17.	The development of technologies is important for the economy.	○	○	○	○	○	○
18.	Technology kills jobs.	○	○	○	○	○	○

Participant ID: P _____ This questionnaire takes about 5 minutes.

Technology Acceptance Questionnaire (TAM)

This questionnaire gives you an opportunity to rate how useful and how easy to use the **SIGHT VR system** is. As much as possible, think about all the tasks you did with the system while you answer. Please read each statement and mark **one circle** to show how strongly you agree or disagree with it. Read the statements carefully, but don't spend a lot of time on any one item; your first impression is fine. You may leave blank any statement you do not want to answer.

		1 Extremely disagree	2 Moderately disagree	3 Slightly disagree	4 Neutral	5 Slightly agree	6 Moderately agree	7 Extremely agree
1.	Using the SIGHT VR system would enable me to learn machine tasks more quickly than other training methods.	◯	◯	◯	◯	◯	◯	◯
2.	Using the SIGHT VR system would improve my performance in my work or studies.	◯	◯	◯	◯	◯	◯	◯
3.	Using the SIGHT VR system would increase my productivity in my work or studies.	◯	◯	◯	◯	◯	◯	◯
4.	Using the SIGHT VR system would make me more effective when working with machines.	◯	◯	◯	◯	◯	◯	◯
5.	Using the SIGHT VR system would make it easier to work safely with machines.	◯	◯	◯	◯	◯	◯	◯
6.	I would find the SIGHT VR system useful in my work or studies.	◯	◯	◯	◯	◯	◯	◯
7.	Learning to operate the SIGHT VR system was easy for me.	◯	◯	◯	◯	◯	◯	◯
8.	I found it easy to get the SIGHT VR system to do what I wanted it to do.	◯	◯	◯	◯	◯	◯	◯
9.	My interaction with the SIGHT VR system was clear and understandable.	◯	◯	◯	◯	◯	◯	◯
10.	I found the SIGHT VR system to be flexible to interact with.	◯	◯	◯	◯	◯	◯	◯
11.	It was easy for me to become skillful at using the SIGHT VR system.	◯	◯	◯	◯	◯	◯	◯
12.	I found the SIGHT VR system easy to use.	◯	◯	◯	◯	◯	◯	◯

Thank you. Please hand this form back to the research assistant.